

Diffie-Hellman Key Exchange Algorithm

Aim:

To implement Diffie-Hellman Key Exchange algorithm in C++.

Algorithm:

Key Generation & Handshake:

* Define public parameters

- Prime $p = 1000000007$

- Generator $g = 5$

* Client computes:

- Private Key a

- Public Key $A = g^a \mod p$

* Server computes

- Private Key b

- Public Key $B = g^b \mod p$

Shared Secret Computation

* Client: $K = B^a \mod p$

* Server: $K = A^b \mod p$

Message Transmission

* Client derives AES Key/IV from K .

* Encrypts message using DH cipher: $encrypt()$

and sends it.

* Server ~~en~~ decrypts using DH cipher: decrypto

Output:

Server:

Listening on port 8080...

Received client public Key A : 481920481

Sent public key B : 294018239

Computed shared secret Key : 719284012

Received 64 bytes of encrypted ciphertext.

Decrypted Message : "Secret message payload

protected by Diffie-Hellman & AES!

Client:

Sent public key A : 481920481

Received Public Key B : 294018239

Computed Shared Secret Key : 719284012

Plaintext : "Secret message payload protected
by Diffie-Hellman & AES!"

Encrypted ciphertext successfully sent to server.